F422 Quantitative Methods in Forest Resource Management

Homework 4: Model I and II Formulation (Due 11/10/2010)

Consider a situation in which there are two initial stands, one stand will be 10 years old in the first planning period and is fully stocked, and another type will be 40 years old in period one and is poorly stocked.

1. Assume planning periods are 10 years in length.
2. Assume that all stands are regenerated as fully stocked stands.
3. Ignore the time value of money by using a discount rate of i = 0%
4. Currently there are 1000 acres of the fully stocked stands and 5000 acres of the poorly stocked stands.
5. Minimum harvest age is 40 years for poorly stocked stands and 30 years for fully stocked stands.
6. All activities occur at beginning of planning interval.
7. Table 1 summarized all technically feasible management alternatives. Each row represents a feasible alternative.

Seq #	Poorly Stocked, Age 40							Seq #	Fully Stocked, Age 10						
	Yield in Period						Net revenue		Yield in Period						Net revenue
	1	2	3	4	5	6			1	2	3	4	5	6	
1							0	1							0
2	9						3	2			11				4
3	9			11			7	3			11			11	8
4	9				18		12	4				18			8
5	9					21	13	5					21		10
6		13					5	6						24	12
7		13			11		9								
8		13				18	13								
9			16				7								
10			16			11	11								
11				19			9								
12					22		11								
13						25	13								

Define the decision variables and write out both the model I (50 points) and II (50 points) formulations based on the information given in table above. The landowner's objective is to maximize the total net revenue from manage this forest within the next 60 years. Both formulations should include the following constraints:

a. constraints for initial available areas of each forest type
b. bookkeeping constraint to track the volume harvested during each period
c. non-declining even flow constraints for timber harvesting
d. non-negativity constraints for decision variables

The model II formulation should also include the bookkeeping constraints for "area harvest = area regenerated" during each period.